Figures for ICML Response

Table 1. Comparison of attack strategies and their properties.

	Voting (greedy) *[prev work]*	**Voting (heuristic)**	**Prompting**	**Nomination**	**Prompting + nomination**
Max rank boost	20 places ↑	15 places ↑	15 places ↑	15 places ↑	44 places ↑
Must identify models?	All models	Only target model A	–	–	–
Cost of Elo calculations (m models n votes; $n \gg m$)	$O(n)$	$O(n)$	0	$\leq O(m)$; depends on desired # of prompt subsets	$O(m)$
Possible defenses	Block bots & prompting w/o voting Harmonic Borda count	Block bots & prompting w/o voting Harmonic Borda count	Stratify/cluster prompts	Harmonic Borda count	Harmonic Borda count Stratify/cluster prompts

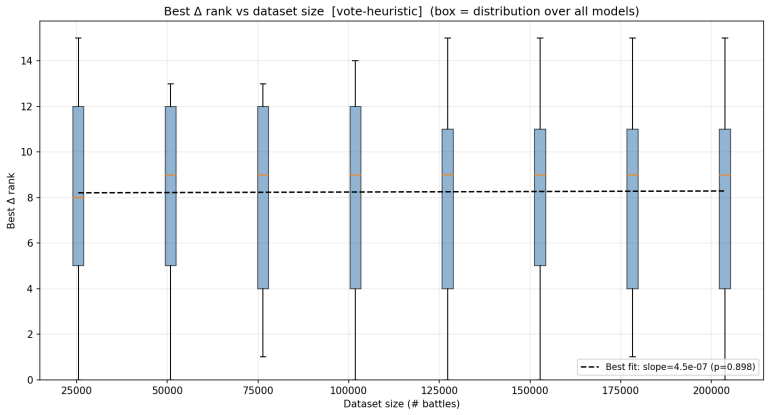

Figure 1. Best delta rank achieved when manipulating a constant 10% of battles on datasets of different sizes, on the *heuristic voting* strategy. When plotting dataset size (x) against best rank achieved with $\leq$ 10% battles modified (y), the slope is near-constant with p >> 0.05 (not significantly different from a line with slope 0). This indicates that the number of manipulations needed to achieve similar rank boosts scales linearly with dataset size.

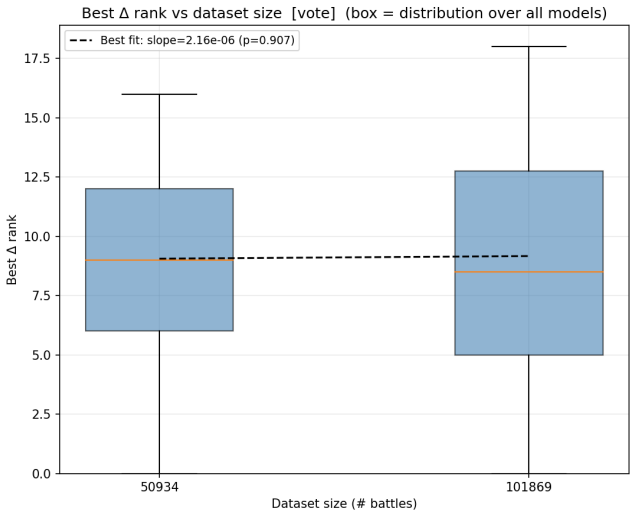

Figure 2. Best delta rank achieved when manipulating a constant 10% of battles on datasets of different sizes, on the *greedy voting* strategy. When plotting dataset size (x) against best rank achieved with $\leq$ 10% battles modified (y), the slope is near-constant with p $>>$ 0.05 (not significantly different from a line with slope 0). This indicates that the number of manipulations needed to achieve similar rank boosts scales linearly with dataset size.

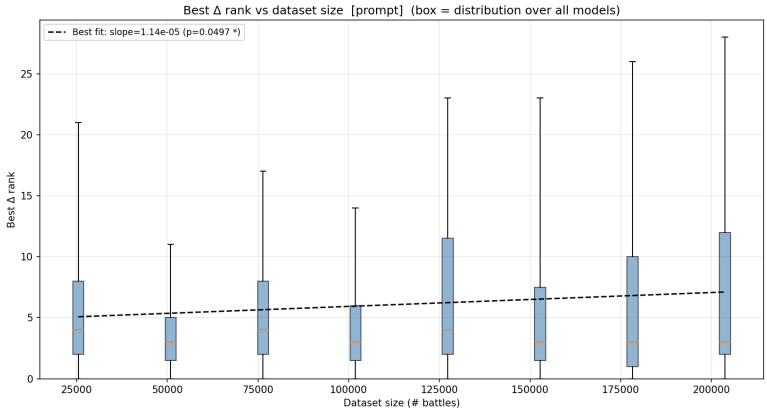

Figure 3. Best delta rank achieved when manipulating a constant 10% of battles on datasets of different sizes, with the *strategic prompting* strategy. When plotting dataset size (x) against best rank achieved with $\leq$ 10% battles modified (y), the slope is very slightly positive, meaning that the manipulation becomes slightly easier as dataset size increases.

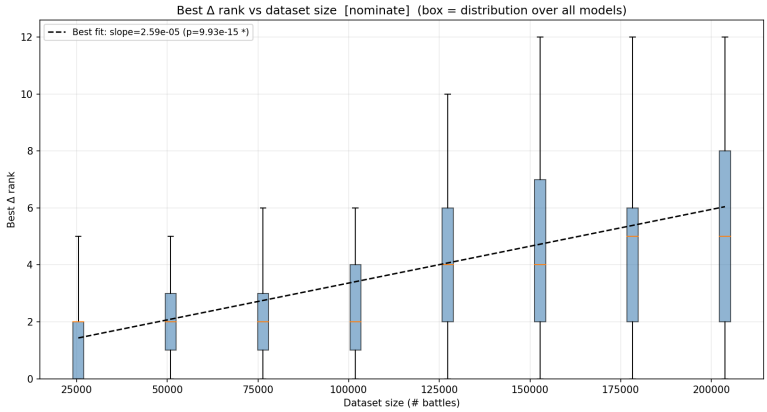

Figure 4. Best delta rank achieved when manipulating a constant fraction of models on datasets of different sizes, using the *strategic nomination* strategy. When plotting dataset size (x) against best rank achieved (y) with ≤ 10 model copies inserted (number of battles per model copy thus scales with dataset size), the slope is slightly positive, meaning the attack is somewhat easier as dataset size increases with the same proportion of models changed (though note that the proportion of models to total votes may vary by leaderboard).

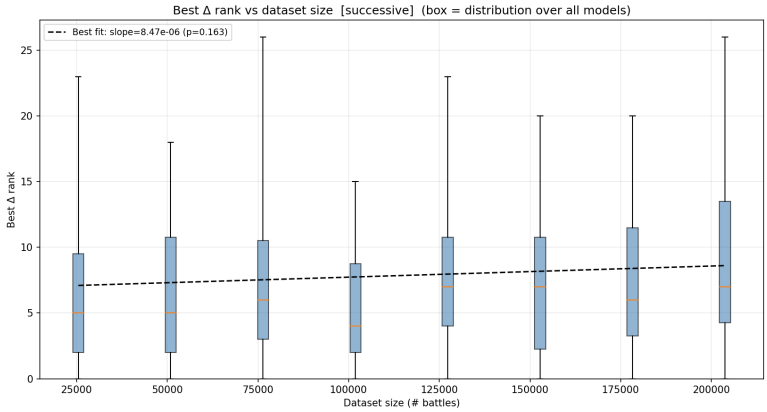

Figure 5. Best delta rank achieved when manipulating a constant 10% of battles on datasets of different sizes, when combining the strategic nomination and prompting strategies in succession. When plotting dataset size (x) against best rank achieved with $\leq 10\%$ battles modified (y), the slope is near-constant with $p >> 0.05$ (not significantly different from a line with slope 0). This indicates that the number of manipulations needed to achieve similar rank boosts scales linearly with dataset size.

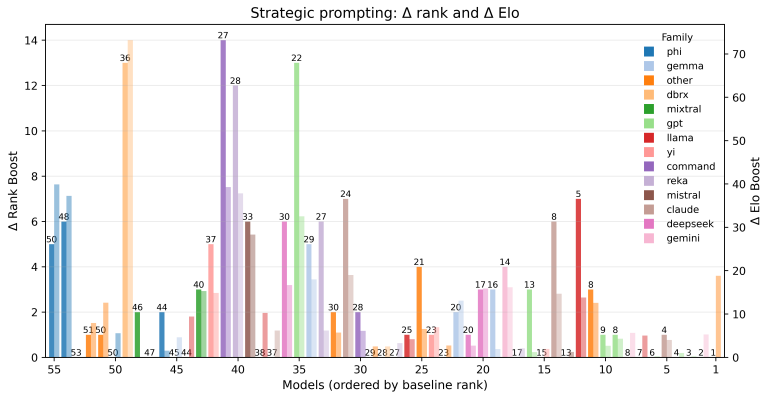

Figure 6. Proof-of-concept that strategic prompting can use any definable, consistent subcategories: best delta rank and Elo achieved achieved under strategic prompting, where prompt categories are based on whether the prompt is labeled as involving complexity, creativity, domain knowledge, and/or math. Rank boosts (up to 14 places) are comparable to those from strategic prompting using language for prompt subcategories.

Table 2. Revised regression coefficients for the best Δ rank achieved from strategic voting.

Feature	Coefficient
elo density vs above	0.354919
winrate variance	0.079747
# total battles	-0.064431
elo gap vs model above	0.028544
winrate vs model above	-0.027852
model Elo	0.026598
# battles vs model above	-0.021019
tie rate	-0.015949
# battles in best prompt lang	-0.011382
model rank	-0.009996
family size	-0.009464
Train R^2	0.6597

Table 3. Revised regression coefficients for the best Δ rank achieved from strategic prompting.

Feature	Coefficient
# battles in best prompt lang	-0.337490
elo gap vs model above	-0.126945
elo density vs above	0.111206
winrate variance	0.069611
family size	-0.066190
winrate vs model above	0.040990
# battles vs model above	-0.039253
tie rate	0.036262
model rank	0.027060
# total battles	-0.016139
model Elo	-0.012417
Train R^2	0.5177

Table 4. Revised regression coefficients for the best Δ rank achieved from strategic nomination.

Feature	Coefficient
elo density vs above	0.231206
elo gap vs model above	-0.200973
winrate variance	0.179882
model Elo	0.157270
# battles in best prompt lang	0.140642
tie rate	0.089433
model rank	-0.077523
# battles vs model above	0.073610
# total battles	-0.036676
family size	0.017748
winrate vs model above	0.001865
Train R^2	0.5572